Comparing and Rounding Decimals

Line up the points. Compare from the left.

0.45 or 0.405?

thousands	hundreds	tens	ones		tenths	hundredths	thousandths
			0	.	4	5	
			0	.	4	0	5

Ones: 0 and 0. Same.
Tenths: 4 and 4. Same.
Hundredths: 5 and 0. 5 is more.

0.45 > 0.405

Round 3.47 to the nearest tenth



The tenth below is 3.4.
The tenth above is 3.5.
3.47 is closer to 3.5.

3.47 rounds to 3.5

Round 6.8 to the nearest whole number



Below: 6. Above: 7.
6.8 is closer to 7.
Right in the middle? Round up.

Your turn!

0.6 > 0.58

For teachers

Goal: compare decimals to thousandths and round them (5.NBT.3b, 5.NBT.4). Say "Line up the points; the first place that is different decides." For rounding: "What is below? What is above? Which one is closer?"

Comparing and Rounding Decimals: Practice

Line up the points. Below, above: which one is closer?

Part 1 Compare in the chart Look for the first place that is different.

1)

th	h	t	o		tth	hth	thth
			2	.	3	5	
			2	.	4		

2.35 < 2.4

2)

th	h	t	o		tth	hth	thth
			0	.	7	1	8
			0	.	7	2	

0.718 < 0.72

3)

th	h	t	o		tth	hth	thth
			5	.	0	6	
			5	.	0	5	9

5.06 > 5.059

4)

th	h	t	o		tth	hth	thth
			0	.	9		
			0	.	9		

0.9 = 0.900

5)

th	h	t	o		tth	hth	thth
		1	4	.	2		
			1	.	4	2	

14.2 > 1.42

Part 2 Compare No help this time. Write <, > or =.

6) 0.8 > 0.79

7) 3.125 < 3.15

8) 0.6 = 0.60

9) 4.08 < 4.8

10) 0.237 < 0.24

11) 7.5 > 7.499

Comparing and Rounding Decimals: Practice (continued)

Part 2 Compare (continued)

12) $1.03 > 1.003$	13) $0.55 > 0.505$	14) $9.1 = 9.10$
15) $2.46 > 2.064$	16) $0.312 < 0.32$	17) $6.07 < 6.7$
18) $0.9 > 0.899$	19) $5.204 < 5.24$	20) $8.3 = 8.300$
21) $0.71 > 0.17$	22) $3.06 > 3.059$	23) $0.402 < 0.42$

Comparing and Rounding Decimals: Practice (continued)

Part 2 Compare (continued)

24) 12.5 [>] 12.49

25) 0.09 [<] 0.9

26) 4.53 [=] 4.530

27) 7.019 [<] 7.09

28) 0.635 [>] 0.6

29) 2.7 [<] 2.71

Part 3 Put them in order No help this time. Smallest first.

30) 0.35, 0.305, 0.53 — [0.305] < [0.35] < [0.53]

31) 2.1, 2.09, 2.15 — [2.09] < [2.1] < [2.15]

32) 4.07, 4.7, 4.017 — [4.017] < [4.07] < [4.7]

33) 0.6, 0.66, 0.606 — [0.6] < [0.606] < [0.66]

34) 1.5, 1.45, 1.405 — [1.405] < [1.45] < [1.5]

35) 3.28, 3.208, 3.3 — [3.208] < [3.28] < [3.3]

Comparing and Rounding Decimals: Practice (continued)

Part 4 Round with the picture Circle the closer one, then write it.

36) 2.6 ← 2.63 → 2.7

2.63 to the nearest tenth = 2.6

37) 7 ← 7.4 → 8

7.4 to the nearest whole number = 7

38) 0.58 ← 0.586 → 0.59

0.586 to the nearest hundredth = 0.59

39) 5.7 ← 5.71 → 5.8

5.71 to the nearest tenth = 5.7

40) 3 ← 3.6 → 4

3.6 to the nearest whole number = 4

41) 1.24 ← 1.243 → 1.25

1.243 to the nearest hundredth = 1.24

Part 5 Fill in below and above Write the one below and the one above. Circle the closer one.

42) 4.8 ← 4.82 → 4.9

4.82 to the nearest tenth = 4.8

43) 9 ← 9.3 → 10

9.3 to the nearest whole number = 9

44) 0.45 ← 0.457 → 0.46

0.457 to the nearest hundredth = 0.46

45) 6.1 ← 6.15 → 6.2

6.15 to the nearest tenth = 6.2

46) 12 ← 12.7 → 13

12.7 to the nearest whole number = 13

47) 2.31 ← 2.318 → 2.32

2.318 to the nearest hundredth = 2.32

Comparing and Rounding Decimals: Practice (continued)

Part 6 Round to the nearest tenth No help this time.

48) 5.38 rounds to 5.4

49) 2.91 rounds to 2.9

50) 7.04 rounds to 7.0

51) 6.66 rounds to 6.7

52) 0.85 rounds to 0.9

53) 3.12 rounds to 3.1

54) 8.25 rounds to 8.3

55) 1.77 rounds to 1.8

Part 7 Round to the nearest whole number No help this time.

56) 8.6 rounds to 9

57) 4.2 rounds to 4

58) 3.5 rounds to 4

59) 11.8 rounds to 12

Comparing and Rounding Decimals: Practice (continued)

Part 7 Round to the nearest whole number (continued)

60) 9.49 rounds to 9

61) 15.3 rounds to 15

62) 6.71 rounds to 7

63) 7.5 rounds to 8

Part 8 Round to the nearest hundredth No help this time.

64) 0.742 rounds to 0.74

65) 1.365 rounds to 1.37

66) 0.218 rounds to 0.22

67) 4.507 rounds to 4.51

68) 2.873 rounds to 2.87

69) 0.691 rounds to 0.69

70) 5.034 rounds to 5.03

71) 0.326 rounds to 0.33

Comparing and Rounding Decimals: Practice (continued)

Part 9 Compare Write <, > or =.

72) $9.32 > 9.119$

73) $4.95 < 6.177$

74) $1.58 < 7.495$

75) $7.24 < 8.718$

76) $3.37 > 0.854$

77) $1.13 > 0.587$

78) $9.65 > 7.427$

79) $6.66 > 4.518$

80) $0.68 < 4.873$

81) $8.37 > 6.177$

82) $5.25 < 6.447$

83) $4.94 > 2.149$

Comparing and Rounding Decimals: Practice (continued)

Part 9 Compare (continued)

84)	$2.65 < 9.298$	85)	$8.94 < 9.952$
86)	$1.43 < 9.432$	87)	$0.88 < 1.836$
88)	$2.98 > 0.351$	89)	$7.58 > 6.667$
90)	$6.29 > 3.357$	91)	$5.59 < 8.875$
92)	$9.79 > 2.521$	93)	$5.23 > 4.715$
94)	$6.76 > 3.299$	95)	$1.83 < 2.969$

Comparing and Rounding Decimals: Practice (continued)

Part 10 Round to the nearest tenth

96) 15.41 rounds to 15.4

97) 7.51 rounds to 7.5

98) 2.12 rounds to 2.1

99) 12.43 rounds to 12.4

100) 0.53 rounds to 0.5

101) 8.52 rounds to 8.5

102) 16.32 rounds to 16.3

103) 17.88 rounds to 17.9

104) 8.66 rounds to 8.7

105) 20.22 rounds to 20.2

106) 0.62 rounds to 0.6

107) 17.72 rounds to 17.7

Comparing and Rounding Decimals: Practice (continued)

Part 10 Round to the nearest tenth (continued)

108) 13.37 rounds to 13.4

109) 8.12 rounds to 8.1

110) 18.87 rounds to 18.9

111) 15.36 rounds to 15.4

112) 18.29 rounds to 18.3

113) 7.86 rounds to 7.9

114) 20.47 rounds to 20.5

115) 6.39 rounds to 6.4

116) 7.54 rounds to 7.5

117) 10.73 rounds to 10.7

118) 2.87 rounds to 2.9

119) 12.64 rounds to 12.6

Comparing and Rounding Decimals: Practice (continued)

Part 11 Remember? Problems from earlier packets.

120) in 6.274 the 7 is worth **0.07**

121) 0.9 = **900** thousandths

122) 3.8 × 1,000 = **3,800**

123) 560 ÷ 100 = **5.6**

124) $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$

125) 4,672 to the nearest hundred = **4,700**

Part 12 Word problems Line up the points.

126) Mia ran 100 m in 15.42 seconds. Leo ran it in 15.4 seconds. Who was faster?

Number sentence **15.4 < 15.42, so Leo took less time**

Answer **Leo**

Draw or show your work

127) A melon weighs 2.68 kg. Round it to the nearest tenth of a kilogram.

Number sentence **2.68 is closer to 2.7**

Answer **2.7** kg

Draw or show your work

128) A ribbon is 1.375 m long. Another is 1.38 m long. Which one is longer?

Number sentence **1.38 > 1.375**

Answer **1.38** m

Draw or show your work

Comparing and Rounding Decimals: Practice (continued)

Part 12 Word problems (continued)

129) A book costs $8.49. About how many dollars is that, to the nearest dollar?

Number sentence **8.49 is closer to 8**

Answer **8** dollars

Draw or show your work

130) Challenge: Three jumps were 3.4 m, 3.09 m and 3.35 m. Write them from shortest to longest.

Number sentence **3.09 < 3.35 < 3.4**

Answer **3.09, 3.35, 3.4** m

Draw or show your work

Check-In: Comparing and Rounding Decimals

Show what you know. No help this time. **Mastery: 10 of 12 (80%)**

Part 1 Compare Write <, > or =.

1) 0.264 **<** 0.3

2) 0.7 **>** 0.69

3) 0.18 **<** 0.8

Part 2 Put them in order Smallest first.

4) 0.42, 0.402, 0.24 → **0.24 < 0.402 < 0.42**

Part 3 Round with the picture

5) **3.7** ← 3.74 → 3.8

3.74 to the nearest tenth = **3.7**

6) **0.36** ← 0.363 → 0.37

0.363 to the nearest hundredth = **0.36**

Part 4 Round to the nearest tenth

7) 5.13 rounds to **5.1**

8) 2.45 rounds to **2.5**

Part 5 Round to the nearest whole number

9) 6.5 rounds to **7**

10) 9.8 rounds to **10**

Part 6 Round to the nearest hundredth

11) 1.624 rounds to **1.62**

12) 3.075 rounds to **3.08**